

Uniform Mutation in a Three-Strategy Rock–Paper–Scissors Flow

HCS Research Program

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Abstract

We give a convention-complete theorem for the cyclic three-strategy flow

$$\dot{x} = ax(y - z) + \mu(1/3 - x), \quad \text{cyclically in } (x, y, z), \quad a, \mu \geq 0,$$

on the unit simplex. At zero mutation the product $H = xyz$ is conserved: every regular interior level is one periodic orbit, the boundary is a cyclic heteroclinic network, and the physical period has an exact turning-point quadrature with center limit $2\pi\sqrt{3}/a$. Positive additive uniform mutation points inward, makes $\log H$ a strict Lyapunov function, and gives global convergence to the barycenter; the tangent rates are $-\mu \pm ia/\sqrt{3}$. We close the exactly solvable $a = 0$ contraction and identity faces and keep finite numerical rows separate from the theorem. This is a source-local population-flow result: there is no arithmetic owner, target determinant, or Hilbert–Pólya claim.

1 Model and simplex geometry

Let

$$\Delta = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \geq 0, x + y + z = 1\}.$$

The vector field is

$$F_{a,\mu}(x, y, z) = (ax(y - z) + \mu(1/3 - x), ay(z - x) + \mu(1/3 - y), az(x - y) + \mu(1/3 - z)). \quad (1)$$

The cyclic terms cancel, hence $\dot{x} + \dot{y} + \dot{z} = 0$. On a coordinate face and $\mu = 0$ the face is invariant. If $\mu > 0$, a zero coordinate has derivative $\mu/3$, so every trajectory enters the open simplex immediately. The barycenter $b = (1/3, 1/3, 1/3)$ is fixed for all parameters.

Theorem 1 (Conservative product and periodic levels). *For $\mu = 0$, $H = xyz$ is constant. The barycenter is the only interior equilibrium. Every level $0 < h < 1/27$ is a single regular periodic orbit; as $h \downarrow 0$ its closure tends to the three-edge heteroclinic network.*

Proof. On the interior,

$$\frac{d}{dt} \log H = a[(y - z) + (z - x) + (x - y)] = 0. \quad (2)$$

The equations $y + z = 1 - x$ and $yz = h/x$ have two distinct roots except at the barycenter. Consequently the compact positive level is a one-dimensional regular component. The cyclic orientation of (1) gives a nonvanishing vector field on it, hence one periodic orbit. Setting one coordinate to zero gives the invariant edges and their cyclic connections; these are the $h = 0$ boundary limit. $\square$

2 Exact physical period

From the two-root relation,

$$(y - z)^2 = (1 - x)^2 - \frac{4h}{x}.$$

Let $x_- < 1/3 < x_+$ be the physical roots of

$$x(1 - x)^2 = 4h, \quad 0 < h < 1/27. \quad (3)$$

The two signs of $y - z$ traverse the closed level, so its exact period is

$$T(h) = \frac{2}{a} \int_{x_-}^{x_+} \frac{dx}{x \sqrt{(1 - x)^2 - 4h/x}}. \quad (4)$$

The cubic has a third root $x_3 = 2 - x_- - x_+ > 1$. With

$$x(\theta) = \frac{x_- + x_+}{2} + \frac{x_+ - x_-}{2} \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},$$

the endpoint cosine cancels exactly and (4) becomes

$$T(h) = \frac{2}{a} \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sqrt{x(\theta)[x_3 - x(\theta)]}}. \quad (5)$$

This regular integral is the deterministic representation used by the ledger.

Proposition 1 (Center and boundary limits). *For $a > 0$, $T(h) \rightarrow 2\pi\sqrt{3}/a$ as $h \uparrow 1/27$, while $T(h) \rightarrow +\infty$ as $h \downarrow 0$.*

Proof. At the center, both physical roots approach $1/3$ and x_3 approaches $4/3$; dominated convergence in (5) gives the displayed constant. At the lower endpoint, the factor $x(\theta)^{-1/2}$ develops a divergent integral, which is the familiar slowing near the heteroclinic edges. $\square$

3 Uniform mutation and convergence

Theorem 2 (Strict product Lyapunov law). *For $\mu > 0$ and every interior point,*

$$\frac{d}{dt} \log(xyz) = \frac{\mu}{3} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 9 \right) \geq 0, \quad (6)$$

with equality exactly at b . Every solution from Δ converges to b ; there is no nonconstant recurrent trajectory.

Proof. The reciprocal AM–HM inequality and $x + y + z = 1$ give $1/x + 1/y + 1/z \geq 9$, with equality only when all coordinates are equal. A boundary initial condition enters the interior for positive time, after which $H > 0$ and (6) applies. The compact positively invariant simplex supplies a nonempty omega-limit set. LaSalle’s invariance principle puts it in the zero-derivative set, which is the singleton $\{b\}$. $\square$

At b , the ambient Jacobian has characteristic polynomial

$$(\lambda + \mu) \left((\lambda + \mu)^2 + \frac{a^2}{3} \right).$$

Thus the tangent-plane eigenvalues are $-\mu \pm ia/\sqrt{3}$. The imaginary part is the residual cyclic rotation; the negative real part is supplied only by the uniform mutation term.

4 Degenerate faces and finite receipt

When $a = 0$, the three coordinates decouple:

$$x_i(t) = \frac{1}{3} + (x_i(0) - \frac{1}{3})e^{-\mu t}. \quad (7)$$

When $a = \mu = 0$, (1) is the identity on all of Δ . For $\mu = 0, a > 0$, the edges remain invariant and the interior period family above is the whole recurrence statement. We make no extension to a general mutation matrix.

The receipt fixes three conservative rates, five rational product levels, three center-limit rows, six mutation starts (including all three coordinate boundaries), three exact contractions, and four tangent spectra. Fixed-step RK4 rows are explicitly labelled diagnostics. The independent checker does not import the producer; SymPy checks sixteen identities, byte replay uses two fresh temporary trees, and the hostile suite rejects 24/24 mutations.

References

- [1] R. M. May and W. J. Leonard, Nonlinear aspects of competition between three species, *SIAM Journal on Applied Mathematics* 29, 243–253 (1975). This citation supplies cyclic-competition context only; the frozen model here is the additive-uniform-mutation replicator field.
- [2] J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics*, Cambridge University Press (1998).
- [3] J. P. LaSalle, Some extensions of Liapunov’s second method, *IRE Transactions on Circuit Theory* 7 (1960).

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, determinant-matching, or Hilbert–Pólya claim. **Data and code.** All rows and audits accompany HCS-C235. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checks displayed formulas and metadata.